DHEERAJ AMARAVADI

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PROFESSIONAL SUMMARY

Applied AI Engineer with hands-on experience building production-oriented AI systems, enterprise AI systems, RAG-based knowledge retrieval, LLM-powered workflows, MCP servers and backend services. Experienced in translating engineering requirements into reusable Agent Skills, integrating enterprise knowledge and capabilities with AI agents and building scalable backend services for agent orchestration and tool execution. Strong software engineering background in Python, C#, JavaScript, REST APIs, with additional experience in machine learning, deep learning, prompt engineering and LLM evaluation. Experienced in applying specification-driven development and collaboration with cross-functional teams to deliver AI solutions for real world engineering workflows.

WORK EXPERIENCE

Artificial Intelligence Engineer | Tektronix | April 2025 – Present

- Develop production-oriented Generative AI application and reusable AI Agent Skills, contributing to enterprise AI platform capabilities.
- Implemented RAG-based knowledge retrieval by structuring application and backend knowledge using Open Knowledge Format (OKF) and integrating contextual knowledge into reusable Agent Skills.
- Apply context engineering and metadata-driven knowledge organization to help agents identify and retrieve relevant enterprise information during task execution.
- Build reusable Model Context Protocol (MCP) servers to expose enterprise tools and capabilities to AI agents.
- Design backend components for AI agent orchestration, tool execution, knowledge retrieval and enterprise integrations.
- Develop LLM-powered workflow automation to streamline engineering processes and reduce manual intervention.
- Apply Specification-Driven Development using GitHub Spec Kit to translate requirements into structured specifications, implementation plans and production-oriented AI applications.
- Collaborate with cross functional engineering teams to translate requirements into maintainable AI solutions and integrate agent capabilities into existing workflows.

Junior Developer | ILaqa Technologies (Remote) | July 2024 – February 2025

- Sole backend developer for an on-demand home services platform, responsible for building and delivering all backend services to production.
- Built 15+ RESTful APIs using Node.js and Express, implementing business logic including service listings with area-based filtering and pagination, referral management and service request status flow management.
- Designed MongoDB schemas for users, services, areas and referrals with cross collection lookups to support relational data fetching across the platform.
- Implemented secure authentication using AWS Cognito with role-based access control, JWT-based endpoint protection and token refresh logic, managed profile AWS S3 image uploads and built production-grade backend infrastructure with centralized error handling, request validation and logging.

AI & LLM Evaluator & Code Reviewer | Outlier.ai (Remote) | May 2024 – June 2024

- Evaluated and ranked LLM-generated code based on correctness, relevance, instruction adherence and overall response quality as part of human-feedback-driven model evaluation (RLHF).
- Designed and refined prompts to elicit accurate, relevant and task-specific model responses.
- Provided structured feedback on model outputs, identifying errors and quality issues to support model improvement and strengthen expertise in LLM evaluation, prompt engineering and human feedback.

ACADEMIC PROJECTS

Crop Recommendation System

- Built a 97% accurate ML model predicting suitable crops based on soil and environmental parameters.
- Utilized Random Forest, Decision Tree, Logistic Regression, and Gaussian Naive Bayes algorithms.
- Tech Stack: Python, Scikit-learn, Flask
  GitHub: https://github.com/AMARAVADIDHEERAJ/Crop-Recommendation-System

Network Intrusion Detection System

- Developed a deep learning-based IDS using 1D CNN and Adam optimizer, balanced imbalanced datasets using SMOTE, achieving 98.59% accuracy.

- Tech Stack: Python, TensorFlow, Pandas, NumPy
- GitHub: https://github.com/AMARAVADIDHEERAJ/Network-Intrusion-Detection-System-Using-Deep-Learning

TECHNICAL SKILLS

AI & Generative AI: Large Language Models (LLMs), AI Agents, Agent Skills, Retrieval Augmented Generation (RAG), Open Knowledge Format (OKF), Model Context Protocol (MCP), LLM Integration, Prompt Engineering, LLM Evaluation
Programming Languages: Python, C#, JavaScript, SQL
AI/ML: Machine Learning, Deep Learning, TensorFlow, Scikit-learn, Neural Networks
Backend & APIs: REST APIs, Node.js, Express.js, API Integration, MongoDB, Microservices, MySQL
Cloud & Engineering: AWS Cognito, AWS S3, Postman, WPF, Object-Oriented Programming, Specification-Driven Development, GitHub Spec Kit

CERTIFICATIONS

- Oracle Cloud Infrastructure 2024 Generative AI Professional | Oracle | certification
- Intermediate Machine Learning (Kaggle Certification) | certification

EDUCATION DETAILS

B. Tech | CSE with specialization in Artificial Intelligence and Machine Learning | 2020 - 2024
Vellore Institute of Technology, Chennai | **CGPA: 7.76**

LEADERSHIP

- **Secretary - National Service Scheme (NSS) VIT CHENNAI -** Organized and led community wellness initiatives, including health camps and awareness drives, impacting 1000+ individuals and strengthening community engagement.